

ALI NIKKHAH

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SUMMARY

Senior AI Engineer & Technical Lead with 2+ years of experience architecting and leading production AI systems at scale. At Digikala, served as AI Engineering R&D Lead for the enterprise intelligence layer serving 5M+ daily queries: designed the LangGraph multi-agent architecture with hybrid RAG (FAISS + Elasticsearch via RRF), HyDE query expansion, and KAG symbolic reasoning; engineered the vLLM/Triton serving stack with PagedAttention achieving 3.2x throughput gain and 60% cost reduction; supervised multi-tenant Kubernetes deployment (200 pods, 40% GPU cost reduction) isolating vision and language model clusters. At Advanced Analytics Australia, led the edge computer vision deployment across 12 industrial sites: TensorRT/TensorFlow Lite compilation, Triton Inference Server, Kafka streaming for 100+ camera feeds at 30 FPS, achieving 50ms inference and 99.2% uptime. At Turquoise Digital, drove the BNPL credit risk modeling initiative (AUC 0.87, -22% default rate) and engineered a financial agentic chatbot with real-time Whisper/wav2vec2.0 voice transcription and secure function-calling. Proven track record of technical leadership: defining R&D roadmaps, mentoring engineers, establishing ML best practices, and delivering measurable business impact through AI innovation.

TECHNICAL SKILLS

Languages & Runtimes: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
Data Engineering: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
MLOps & Serving: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
DevOps & Infrastructure: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
Databases & Monitoring: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Led a team of 5 engineers architecting the enterprise AI intelligence layer serving over **5 million daily queries**; defined the R&D roadmap transitioning from static semantic matching to stateful LangGraph multi-agent pipelines with hybrid RAG (FAISS + Elasticsearch via RRF), HyDE query expansion, and KAG symbolic reasoning (**+23% recall@10, -41% factual error rate**).
- Designed the vLLM/Triton Inference Server serving stack with PagedAttention, async FastAPI front-end, multi-tier Redis caching, and request batching achieving **3.2x throughput gain, 60% cost reduction per 1K queries, p99 latency 200ms**; established ML engineering best practices: evaluation pipelines, guardrails, A/B testing framework.
- Supervised multi-tenant Kubernetes deployment (200 pods) isolating vision and language model compute nodes for independent horizontal scaling and fault containment (**40% GPU cost reduction**); mentored 3 engineers on agentic systems, LLM serving optimization, and MLOps practices.
- Established cross-functional collaboration with product, data engineering, and infrastructure teams; defined SLIs/SLOs for AI services; implemented automated model evaluation and canary deployment workflows.

Digikala

Tehran, Iran

AI Engineer

Nov 2024 – Jan 2025

- Prototyped and benchmarked RAG architectures against real production query logs; integrated LangGraph cyclic agent graphs for multi-turn session management, dynamic context re-evaluation, and automated query rewriting loops.
- Evaluated retrieval precision and end-to-end latency across candidate retrieval systems; findings directly shaped the R&D roadmap adopted in the subsequent lead role.

Advanced Analytics Australia

Remote Contract

Computer Vision Tech Lead

Sep 2025 – Apr 2026

- Led the edge computer vision deployment across **12 industrial sites** for health, safety, and compliance monitoring: handwash verification, PPE detection (boots, helmets), face tracking, and restricted-zone entry alerting (**50ms inference, 99.2% uptime, -35% false positives**).

- Compiled and quantized vision models into **TensorRT** and **TensorFlow Lite** runtimes; served them through **Triton Inference Server** with dynamic batching (**3.5x speedup vs Python**); architected Kafka-based ingestion broker for **100+ cameras (30 FPS each)**.
- Defined sensor calibration strategy and continuous SNR monitoring to maintain stable detection thresholds across variable indoor/outdoor lighting conditions; established edge MLOps workflow for model updates across distributed sites.
- Built asynchronous Python streaming workers interfacing directly with camera APIs; mentored 2 engineers on edge deployment, TensorRT optimization, and distributed streaming architectures.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer & Data Scientist

Sep 2025 – Mar 2026

- Drove the BNPL credit risk modeling initiative end-to-end: developed ensemble models (AUC 0.87, **-22% default rate**) with automated fund allocation, containment decisions, and full audit logging; presented model performance and business impact to C-suite for \$2M+ portfolio decisions.
- Engineered an enterprise **LangGraph** agentic chatbot handling customer support ticketing, financial literacy instruction, and live portfolio inquiries; integrated real-time **Whisper** and **wav2vec2.0** voice transcription (**-78% manual tickets, 94% transcription accuracy**); implemented secure function-calling for live equity balance queries without exposing database credentials.
- Built the AI portfolio allocation system: time-series transformer models fusing historical pricing with streaming news-sentiment embeddings, producing forecasts with explicit confidence intervals driving automated trading allocations (**Sharpe +0.42 vs benchmark**).
- Architected the central enterprise data lake and OLAP warehouse (Trino, PeerDB CDC **¡60s latency, 5TB/day**); designed Airflow DAGs for multi-stage transformations; refactored PySpark ETL (**60% memory reduction**); implemented Great Expectations for data quality gating.
- Led data engineering best practices: established data contracts, lineage tracking, and observability (Prometheus, Grafana, ELK); mentored junior engineers on distributed data processing and ML model deployment.

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Established the ML platform from scratch: Kubernetes + Ray + MLflow + TensorFlow Serving; automated CI/CD with Docker, ArgoCD, and GitHub Actions (**¡10min releases**); built FastAPI inference APIs; achieved **15+ models in production, 99.9% deploy success rate**.
- Designed distributed training workflows with Ray; implemented experiment tracking, artifact lineage, and model registry with MLflow; trained team on MLOps practices, containerization, and Kubernetes operations.

SELECTED PROJECTS & RESEARCH

Enterprise Agentic AI Platform

Digikala

Technical Lead

2024 – 2025

- Led end-to-end delivery of multi-agent RAG platform at 5M+ daily queries: LangGraph orchestration, hybrid RAG (FAISS + Elasticsearch via RRF), HyDE, KAG, vLLM/Triton serving, multi-tenant K8s — from architecture through production rollout.
- Established AI engineering culture: code review standards, evaluation pipelines, guardrails, A/B testing framework, SLI/SLO definitions, on-call rotations.

Edge Computer Vision Deployment at Scale

Advanced Analytics Australia

Tech Lead

2025 – 2026

- Delivered production edge CV across 12 sites: TensorRT/TF Lite compilation, Triton serving, Kafka streaming (100+ cameras), dynamic sensor calibration, continuous SNR monitoring — achieving ¡50ms inference, 99.2% uptime.
- Built edge MLOps pipeline for zero-downtime model updates across distributed sites; established monitoring and alerting for inference health, drift detection, and hardware failures.

BNPL Credit Risk & Financial AI Platform

Turquoise Digital

Technical Lead

2025 – 2026

- Drove credit risk modeling end-to-end: ensemble models (AUC 0.87, -22% default rate), automated fund allocation/containment, audit logging, C-suite reporting for \$2M+ decisions.
- Engineered LangGraph agentic chatbot with Whisper/wav2vec2.0 voice AI (-78% tickets, 94% accuracy), secure function-calling, time-series transformer portfolio allocation (Sharpe +0.42).
- Architected data platform: Trino + PeerDB CDC (¡60s latency, 5TB/day), Airflow DAGs, PySpark ETL (60% memory reduction), Great Expectations quality gates.

ML Platform Foundation

HomaCloud

Founding MLOps Engineer

2021 – 2022

- Built ML platform from scratch: K8s + Ray + MLflow + TF Serving; CI/CD (ArgoCD, GitHub Actions ¡10min); 15+ models, 99.9% deploy success; trained team on MLOps practices.

Automated Ultrasound Report Generation

ML Researcher

University of British Columbia

Feb 2024 – Nov 2025

- Developed vision-language architectures (ViT + BLIP + T5) with self-supervised contrastive learning for DICOM-compliant ultrasound report generation; achieved SOTA BLEU/ROUGE-L.

Texture-Free Video Motion Classification

Research Assistant

L3S Research Center

Apr 2023 – Feb 2024

- Designed 3D CNN + transformer pipelines over optical flow for texture-free motion classification; achieved SOTA on UCF-101 and HMDB-51.

RESEARCH HIGHLIGHTS

University of British Columbia

Research Assistant – Vision-Language Medical Imaging

Remote Research Collaboration

Feb 2024 – Nov 2025

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

L3S Research Center (Leibniz University Hannover)

Research Assistant – Video Motion Classification

Remote Research Collaboration

Apr 2023 – Feb 2024

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

EDUCATION

Sharif University of Technology

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Tehran, Iran

Sept 2020 – Jul 2024